RAJU JONNADA

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EDUCATION

Gayatri Vidya Parishad College of Engineering | B-Tech — **Madhuruwada**
Information Technology | *CGPA : 8.10* — *2022-2025*

Sanketika Polytechnic College | Diploma — **P.M.Palem**
Computer Engineering |*Percentage:70.43%* — *2018-2021*

NT School | School — **Akkayyapalem**
CGPA:9.3 — *2017-2018*

SKILLS

- **Languages**: Python, C, JavaScript
- **Frontend**: HTML, CSS, TailwindCSS
- **Backend Frameworks**: FastAPI, Flask, Node.js
- **Database**: SQLite, MySQL, PostgreSQL, Supabase
- **Network Tools**: Netcat, cUrl, SSH, Port Forwarding, Domain Configuration
- **Tools**: Git, GitHub, Linux, LLDB, PDB, VS Code, Postman
- **Core Concepts**: LLM, RAG, NGINX, Deployment using Render, Vercel

WORK EXPERIENCE

BHABHA ATOMIC RESEARCH CENTER (B.A.R.C) | Intern — **Visakhapatnam**
Project: Developed a Convex & Concave Hull Model for Defect Analysis in Materials — may24 – Jul 24

- Researched and analyzed various methods for defect analysis in materials.
- Conducted extensive testing and validation to ensure the model's reliability and efficiency.

Achievements:

- Increased the accuracy of defect analysis by **20%,** significantly improving the evaluation process.
- Collaborated with a team of scientists and engineers to integrate the model into existing systems, ensuring seamless adoption and application.

PROJECTS

AI Banking Data Assistant | GitHub | Preview

- Developed a FastAPI backend integrated with **Open Router LLM API to convert natural language queries into SQL queries** with a 10-model fallback chain for fault-tolerant execution.
- **Integrated Sarvam AI for multilingual voice support in Telugu, Hindi, Tamil and other Indian languages with auto language detection and native voice responses.**
- Designed an interactive dashboard with Chart.js visualizations, real-time credit/debit summary cards, dark/light mode theming and one-click CSV/PDF export functionality.
- Deployed production-grade application on **Render with Supabase PostgreSQL, configured SSL-secured database connections**, environment-based secrets.

Codebase AI —AI-Powered Code Intelligence & Developer Assistant | GitHub

- **Built an AI-powered code intelligence system using Retrieval-Augmented Generation (RAG) architecture to enable contextual interaction with entire software projects, ensuring responses are grounded in actual source code and reducing hallucinated outputs.**
- **Developed backend services with FastAPI and integrated PostgreSQL (pgvector) for vector-based semantic search.**
- Designed and implemented a complete code ingestion pipeline including project upload (ZIP/GitHub), source file filtering, intelligent code chunking, embedding generation using transformer-based models, and context-aware LLM response generation.
- Implemented multi-mode system **supporting code explanation, debugging, review, and interview preparation based on real project context.**

Agentic Video Assistant —AI Recording, Transcription Platform| GitHub

- Built an end-to-end **audio/video recording platform with AI-powered transcription and semantic search using FastAPI, React, PostgreSQL, and pgvector.**
- Designed a multi-provider AI service layer supporting **OpenAI, OpenRouter, Groq, Sarvam, and local Whisper,** with intelligent fallback routing for reliability.
- Implemented **semantic search over recordings using vector embeddings with pgvector** (cosine similarity) along with keyword-based fallback for improved retrieval accuracy.
- Developed an **AI agent ecosystem including an MCP server with structured tools (transcribe, summarize, search, Q&A) and a Type-based CLI for automated testing and AI workflows.**

Real-Time Bottle Detection — YOLOv8 Plastic vs Steel Classifications| GitHub

- Developed a **real-time object detection model using YOLOv8 to classify plastic_bottle and steel_bottle, achieving 82.8% mAP@50, 87.3% Precision, and 76.3%** Recall on a custom dataset.
- Built and **annotated a custom dataset of 544 images using Roboflow, applying data augmentation (flip, rotation, brightness) to expand training data to 1106 images across Train/Validation/Test splits.**
- Implemented Transfer Learning with pre-trained YOLOv8s weights on **Google Colab Tesla T4 GPU, optimizing training with AdamW optimizer and Early Stopping to complete 52 epochs in <25 minutes**.
- Deployed the model for **real-time webcam inference using OpenCV and JavaScript browser APIs, enabling live bottle detection with bounding boxes, confidence scores, and class counting.**

ACHIEVEMENTS

- Successfully deployed CityBus Advisor on Render, configuring backend services and public APIs to make the application accessible over the internet | **Preview**
- Participated in the **AI for Vizag Buildathon 2026**, a **24-hour AI innovation hackathon**, where I developed and demonstrated **AI-driven solutions in a collaborative engineering environment**. | Certificate